Technische Hochschule Augsburg

STATISTICS

International Information Systems, 3rd semester

(7) Towards Inferential Statistics: Sampling and Hypothesis Testing

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BEFORE WE GO ON...
LET'S DO A RECAP

THIS IS A GOOD CHANCE TO SEE IF YOU ARE BEHIND & TO ASK QUESTIONS

WEEK 1: INTRODUCTION TO STATISTICS

Hypotheses and Variables

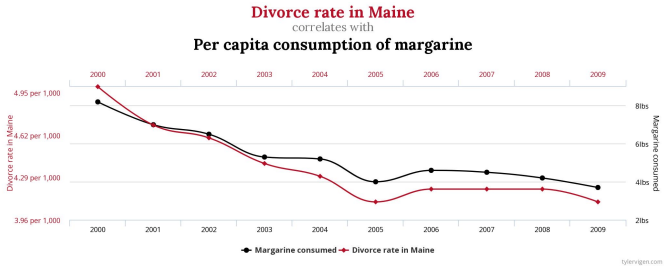
- We have learned how to formulate research questions as ***testable hypotheses*** with ***dependent*** and ***independent variables***
- We know that we need to identify possible **proxies** for a variable

"Eating fast food causes stress"

proposed cause — *proposed outcome*

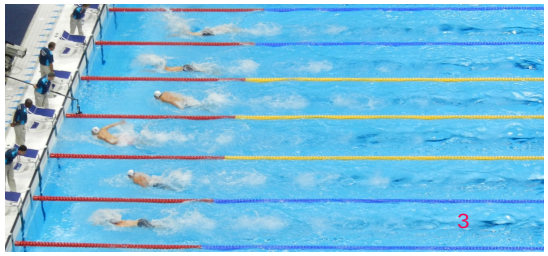
Experimental vs. Correlational research

- We understand and can explain the difference between ***correlational*** and ***experimental research***

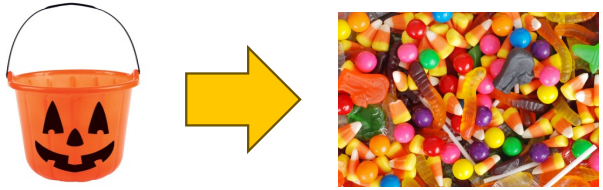


Scales of Measurement

- We can distinguish between ***categorical*** and ***continuous variables***
- We can identify the appropriate ***scale of measurement*** for a variable



WEEKS 2 AND 3: DESCRIPTIVE STATISTICS (1)

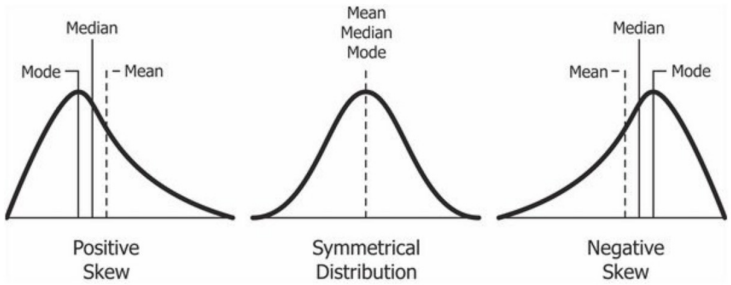


Descriptive and Inferential Statistics Introduction to Descriptive Statistics

- We understand and can explain the difference between ***sample*** and ***population***, ***descriptive*** and ***inferential*** statistics, ***parameters*** and ***statistics***

Frequency Distributions

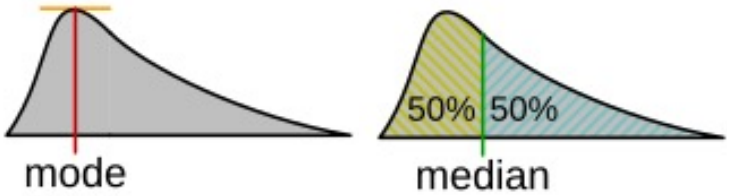
- We can describe and visualize the ***shape*** of the distribution of your data and the properties of the distribution



Measures of Central Tendency

- We can compute the **mode, mean, median** and understand what they tell us about the data (and what they don't)

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$



WEEKS 2 AND 3: DESCRIPTIVE STATISTICS (2)

↗ Measures of Dispersion

- ↗ We can compute **range**, **interquartile range**, **variance**, **standard deviation**, and understand what they tell us about the data

$$s_x = \sqrt{\frac{\sum(X - \bar{X})^2}{N - 1}}$$

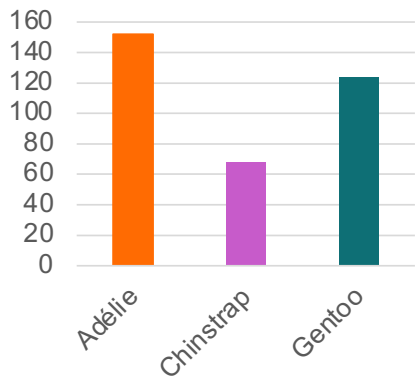
↗ Outliers

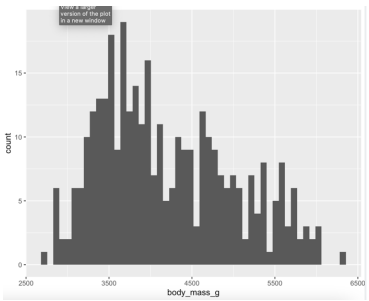
- ↗ We can identify outliers in our data (with **n** z-scores or with the box-and-whiskers method), reflect about them and decide how to handle them

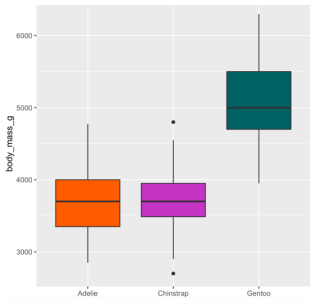
$$\frac{X - \bar{X}}{\mathrm{SD}}$$

↗ Visualizations

- ↗ We can choose the right way to visualize your data and create data visualizations







WEEK 4: INTRODUCTION TO PROBABILITY THEORY

Three views of probability

- We understand and explain the ***analytic view***, the ***frequentist view*** and the ***Bayesian view*** of probability

Independent events

- Computing and manipulating probabilities of independent events

$P(\text{m or m}) = P(\text{m}) + P(\text{m})$

$P(\text{m} \cap \text{m}) = P(\text{m}) \times P(\text{m})$

Conditional probabilities

- Understand and compute the ***accuracy***, ***sensitivity***, ***specificity*** and ***precision*** of a test
- Explain how believes are updated with the ***Bayes rule***

$$P(B|A) = \frac{P\ (A \cap B)}{P(A)}$$

sensitivity: $P(\text{pos_test}|\text{pos_result})$
specificity: $P(\text{neg_test}|\text{neg_result})$
precision $P(\text{pos_result}|\text{pos_test})$

Questions?

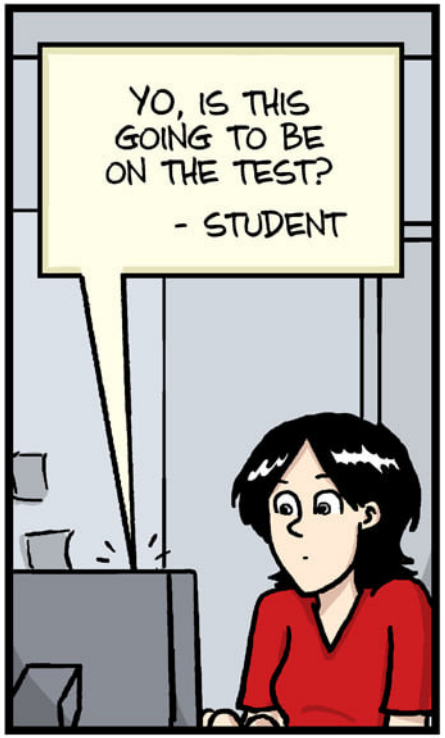
Recap exercise in
R_exercise_8.R
preferences.csv

PRELIMINARY COURSE PLAN

- Introduction, Descriptive statistics: measures of central tendency
- Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations
- Basic probability theory
- **[Oct 30th] Introduction to inferential statistics: sampling, hypothesis testing**
- [Nov 6th] Test Statistics (I)
- [Nov 13th] **No class *(Tag der Lehre)***
- [Nov 20th-Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

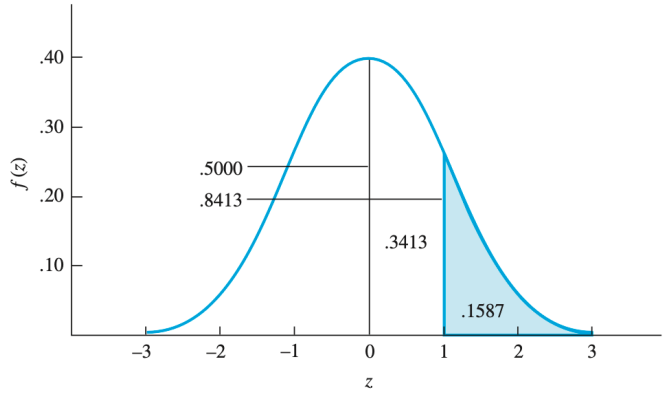
https://phdcomics.com/comics/archive_print.php?comicid=1873

DON'T FORGET TO REGISTER FOR THE EXAM!

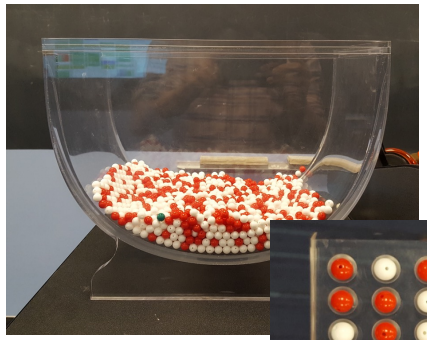


- Register for the exam by Nov 17th!
- Make sure you register for the right "Statistics" exam
  - Exam nr.: 9772010

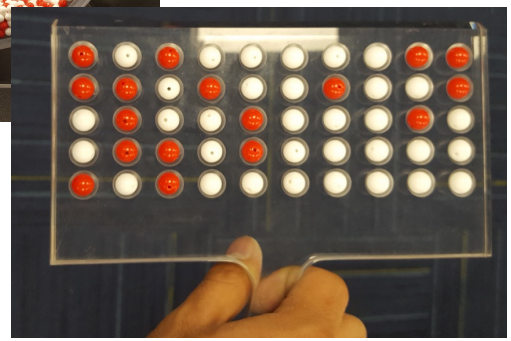
TODAY'S TOPICS AND LEARNING GOALS



- **From Distributions to Probabilities with the Normal Distribution**
  - Compute the area under the curve
  - Interpret the area under the curve as a probability

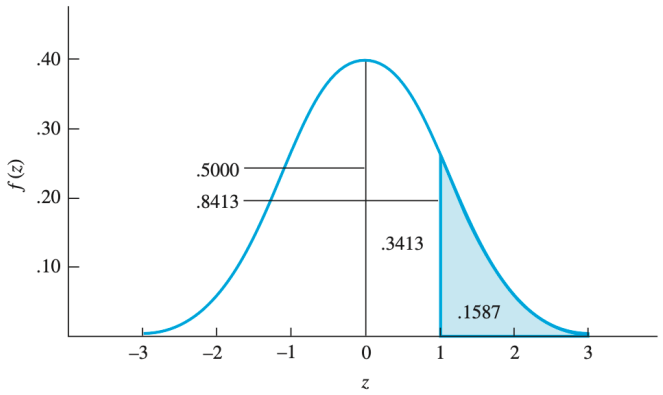


- **Sampling**
  - Explain the Central Limit Theorem and the importance of sample size



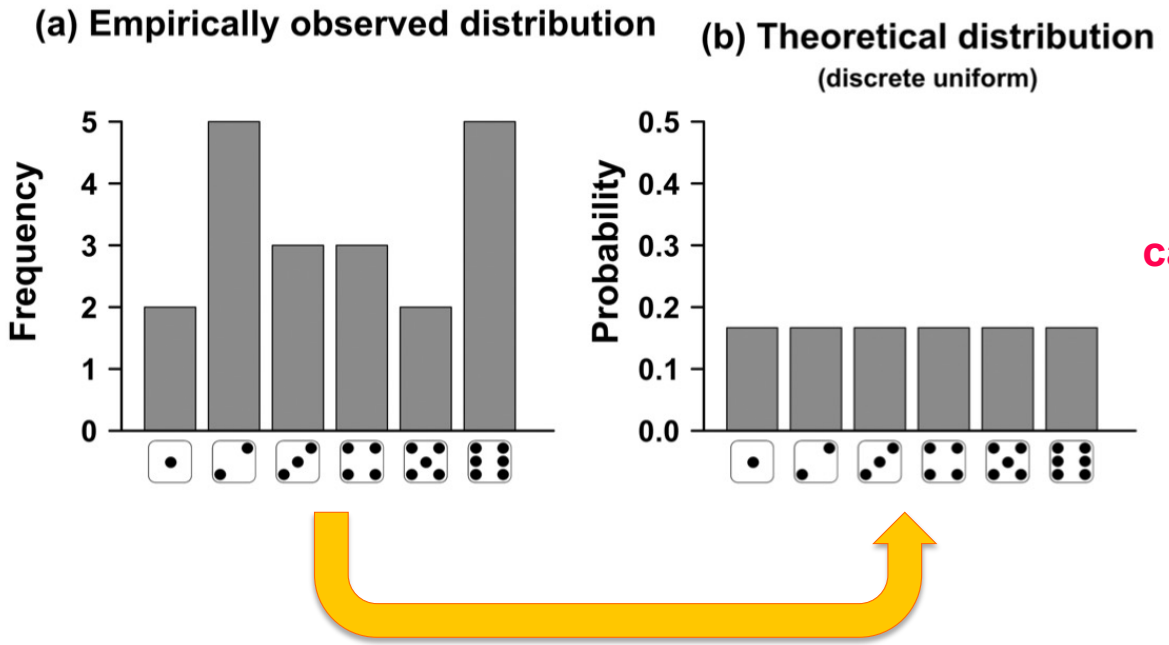
- **Hypothesis Testing with the Normal Distribution: single scores**
  - Formulate the Null Hypothesis for a given hypothesis
  - Test if a score does not come from a population (hypothesis testing on single scores)

FROM DISTRIBUTIONS TO PROBABILITIES WITH THE NORMAL DISTRIBUTION



FROM DISTRIBUTIONS TO PROBABILITIES

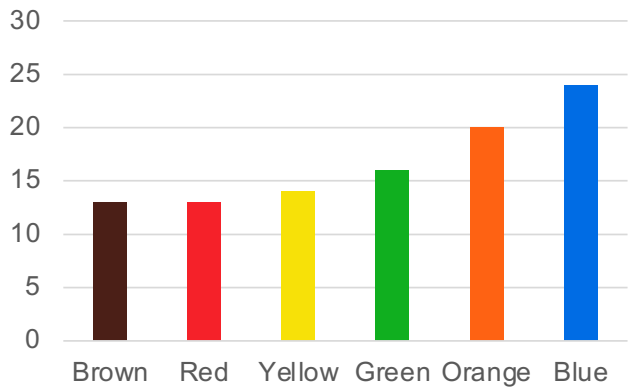
- Distributions can be so informative because they tell us something about the probability of an event (a value that a variable can take)
- The frequency distribution of the values of a variable in a sample can help us ***guess*** the probability of the values in the population (frequentist view)



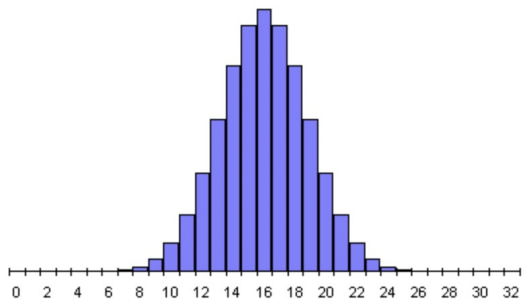
Distribution of a categorical variable

FROM DISTRIBUTIONS TO PROBABILITIES

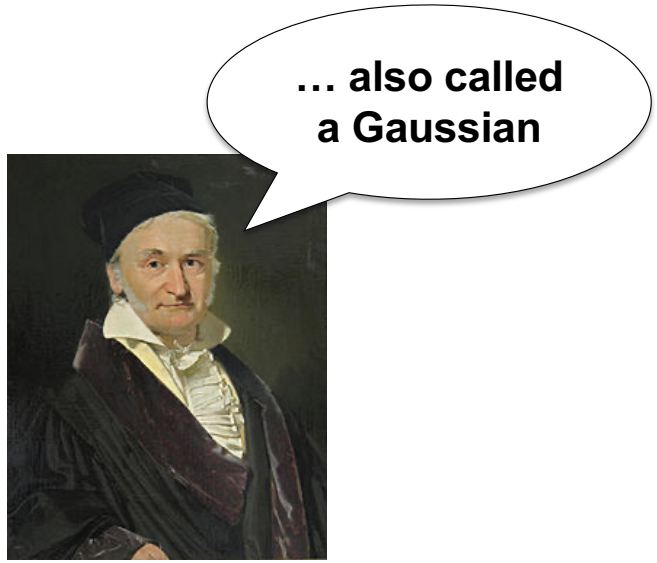
- So far we talked about the probability of ***discrete events***
  e.g. pulling out a blue M&M, rolling 6 on a die…
- But we have also seen distributions of
  continuous variables
  e.g. normally-distributed variables
  (height, amount of rain, sunny days in a year)



How can we use the distribution of a continuous variable to reason about probabilities?



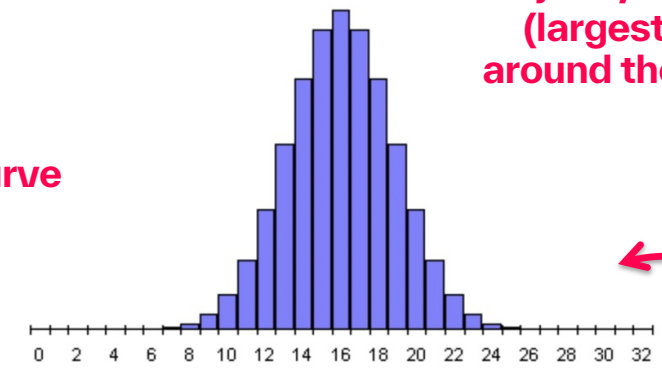
THE NORMAL DISTRIBUTION



- The distribution of data in a **normal distribution** is
  - ***symmetrical*** around the center (not skewed, skew = 0)
  - not flat or pointy (kurtosis = 0)
  - bulk of the data in the middle

majority of scores (largest bars) around the *center*

bell-shaped curve



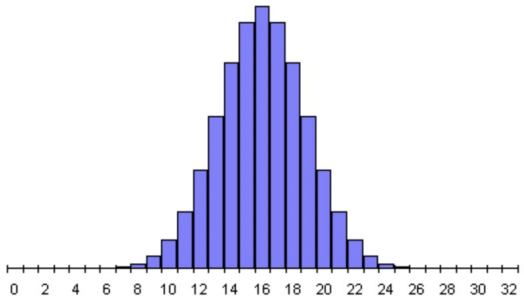
deviating scores are less frequent in the *tails* (smaller bars)

THE NORMAL DISTRIBUTION

↗ **Assumption of normality:**
many statistical procedures assume that
if we were to observe ***the whole population***,
the data of the whole population would be **normally distributed**

Why?
If we can assume that a variable is normally distributed,
we can make a number of **inferences about the population**
starting **from the sample**

↗ Given what we know **about the sample**,
what can we guess / infer / expect
about **the values from the population**?



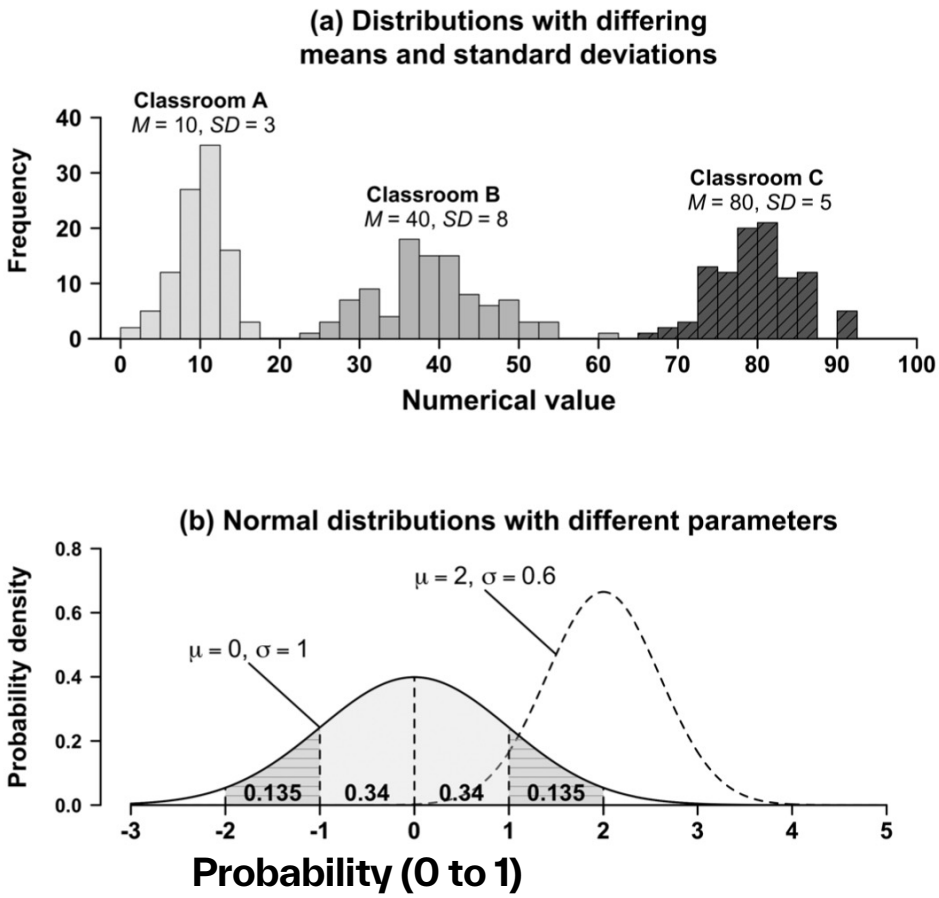
NORMAL DISTRIBUTIONS

- Different normal distributions can differ in terms of
  - their mean (μ)
  - their spread (σ)

Parameters defining the distribution

- A theoretical distribution for continuous data is described using a line

Density plot: a smoothed, "idealized" version of a histogram



THE NORMAL DISTRIBUTION

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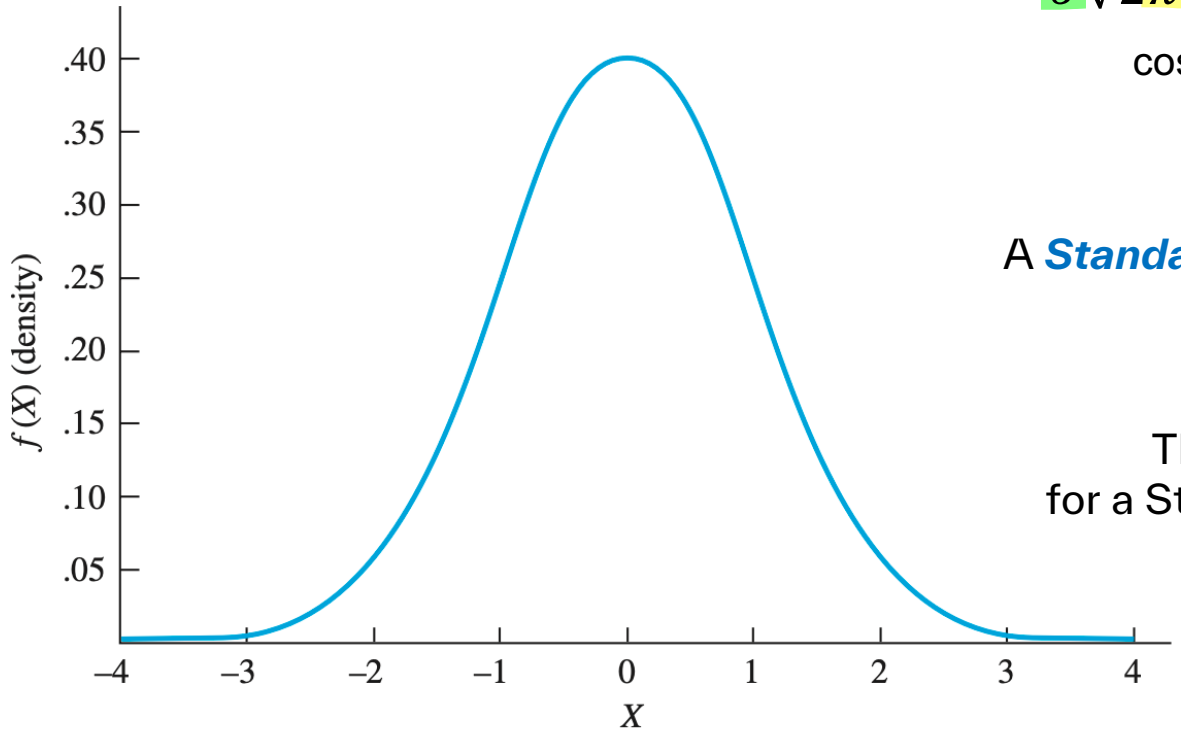
height of the curve for x

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

costants

All you need to define a normal distribution $N(\mu, \sigma^2)$ is μ and σ



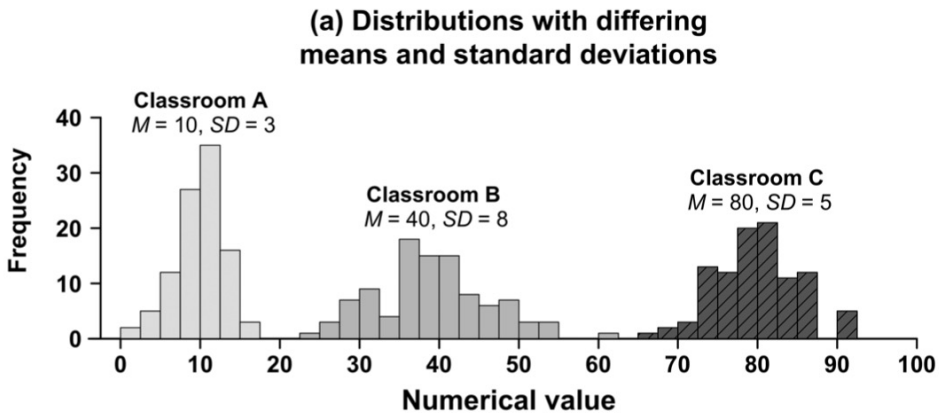
A ***Standard Normal Distribution*** ($N(0,1)$) is defined by

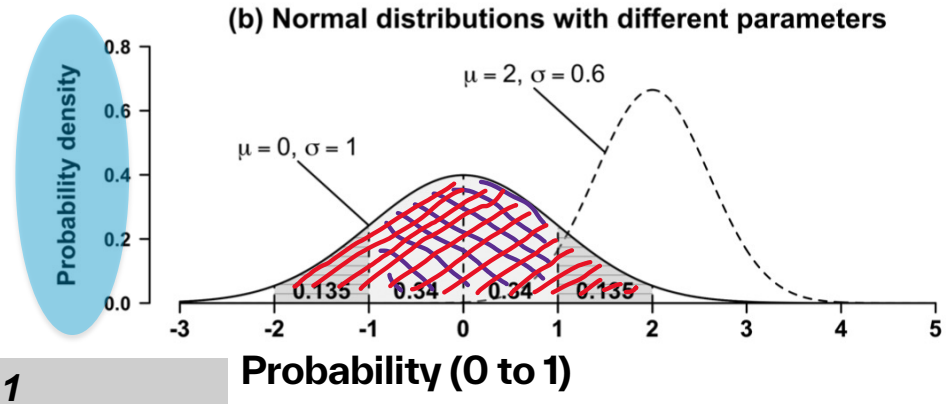
$\mu = 0$ and $\sigma = 1$

The ***Area under the Curve*** for a Standard Normal Distribution is 1

BUT WHAT DOES THIS HAVE TO DO WITH PROBABILITIES?

- Using the parameters of the distribution, we can learn something about the ***probability of the values*** that a variable can take
- A ***probability distribution*** (curved line) is an idealized version of a distribution of scores
- If you were to randomly draw a score from this distribution,
  - 68% of the time you would end up in the **purple area** (1 SDs from the mean)
  - 95% of the time you would end up in the **red area** (2 SDs from the mean)



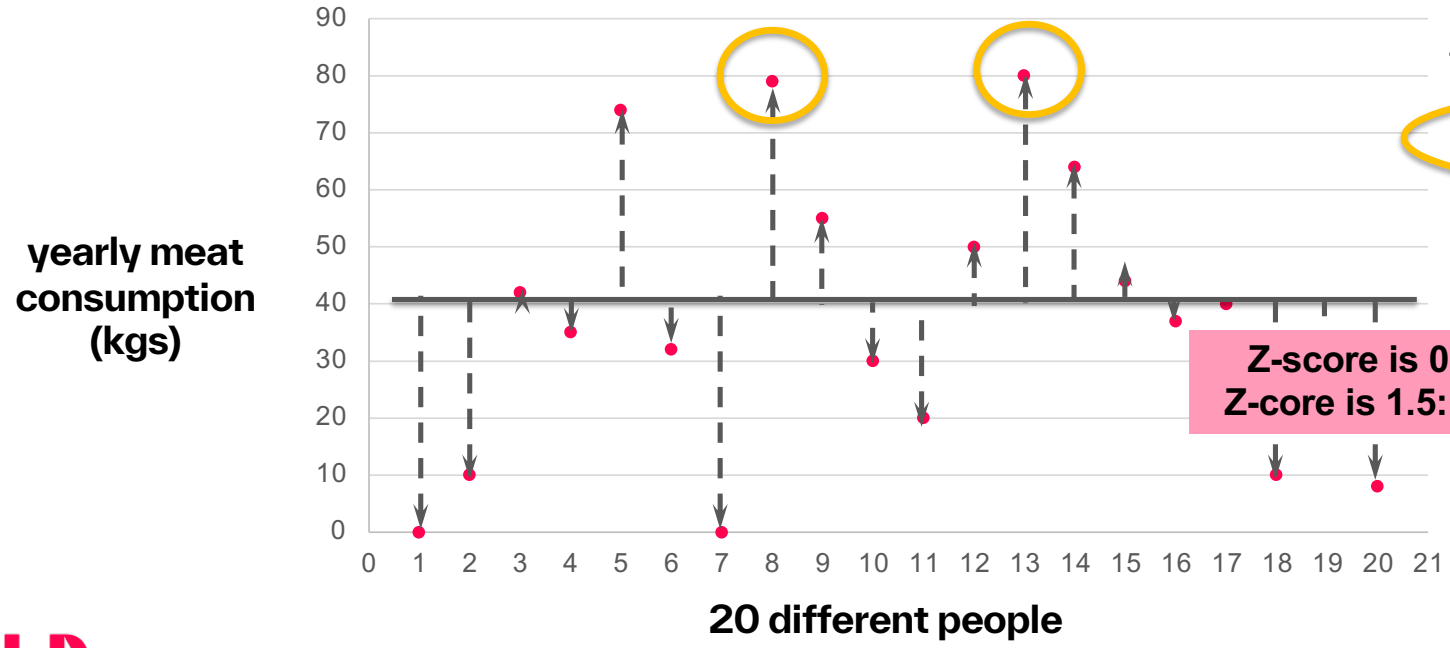


Area under the curve = 1
1 = probability of the variable having one of the values (any) in the range

...REMEMBER THE Z-SCORES?

- A z-score indicates the number of standard deviations that a score is above (positive z-score) or below the mean (negative z-score)

Score n	Score	Z-score
1	0	-1.48
2	10	-1.08
3	42	0.20
4	35	-0.07
5	74	1.48
6	32	-0.20
7	0	-1.48
8	79	1.69
...	...	...



Z-score is 0: we are exactly at the mean
Z-core is 1.5: we are 1.5 SD from the mean

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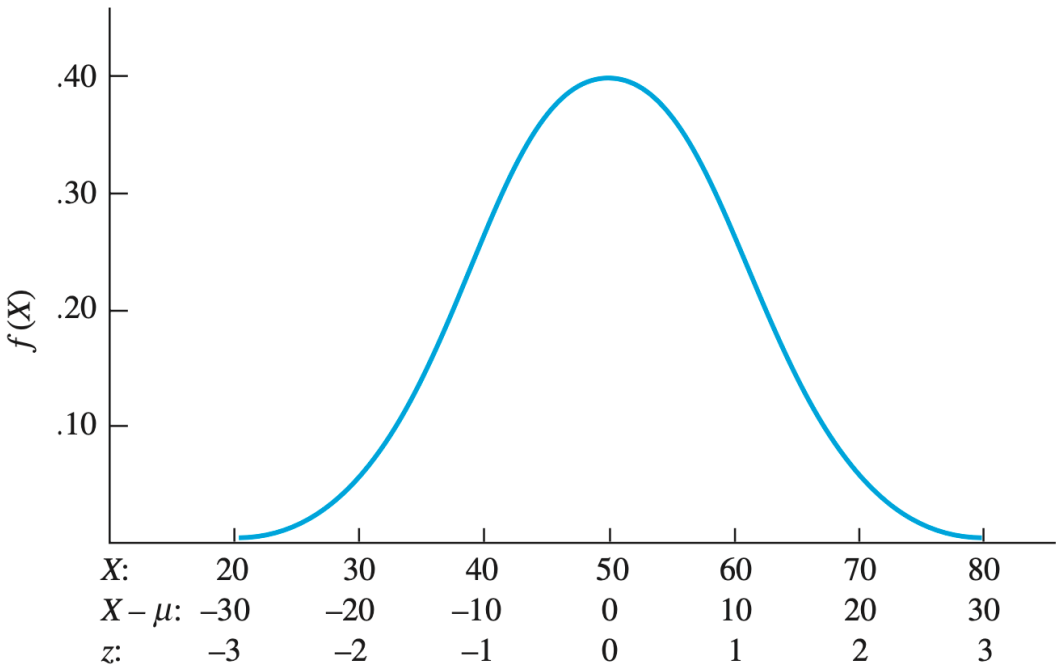
Z-SCORES AND THE NORMAL DISTRIBUTION



- Given a variable X and its mean μ and SD σ, we can transform the variable distribution into a ***Distributions of z-scores*** for X (it's called ***standardization***)
- A distribution of z-scores is a Standard Normal Distribution

$$N(50,100) \Rightarrow N(0,1)$$

- Now we can use what we know about the Standard Normal Distribution (area under the curve = 1) **to talk about probabilities**!

Error (deviation from the mean)

$$z = \frac{(X - \mu)}{\sigma}$$

Error defined in terms of SDs

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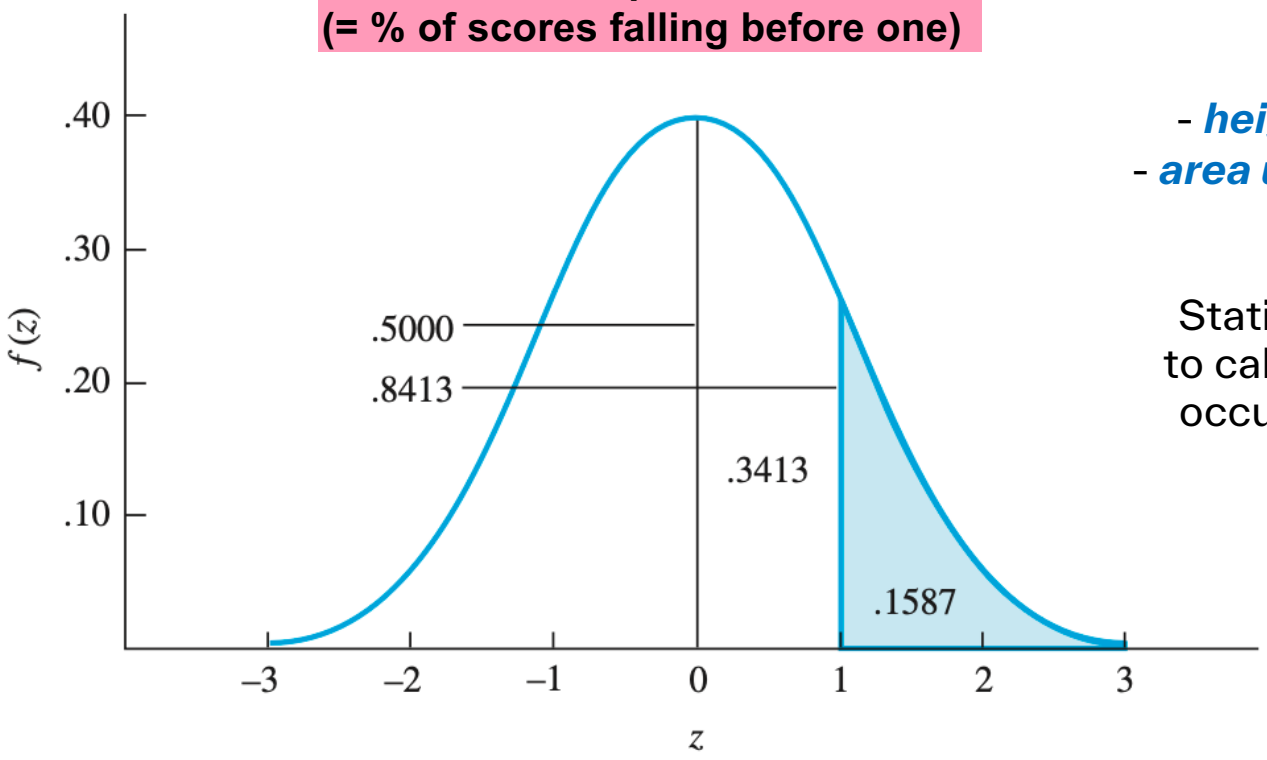
FROM DISTRIBUTIONS TO PROBABILITIES

These are percentiles!
(= % of scores falling before one)

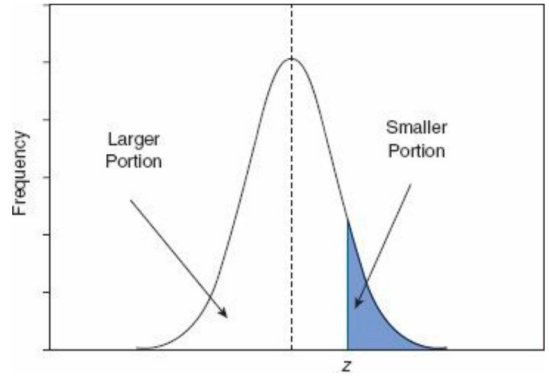


In a standard normal distribution:
- ***height of the curve***: probability of a score
- ***area under the curve***: probability that a score would fall in a certain range

Statisticians use the ***area under the curve*** to calculate the probability of certain scores occurring in a standard normal distribution

What's the probability that if you take a random datapoint from this distribution, it will be > 1 SD higher than the mean? **0.1587**
(that is, it will fall in the blue area)

HOW DO I FIND THE AREA UNDER THE CURVE?



- You can use a Table of Probabilities (from a statistics book)
- ... or you can use R!
  - height of the (normal) curve at x given x, the mean and SD
    `dnorm(x, mean = m, sd =s)`
  - area under the (normal) curve for scores *bigger than* x ("larger portion" above)
    `pnorm(x, mean = m, sd = s)`
  - area under the (normal) curve for scores *lower than* x ("smaller portion" above)
    `1 - pnorm(x, mean = m, sd = s)`

z	Larger Portion	Smaller Portion	y
.00	.50000	.50000	.3989
.01	.50399	.49601	.3989
.02	.50798	.49202	.3989
.03	.51197	.48803	.3988
.04	.51595	.48405	.3986
.05	.51994	.48006	.3984
.06	.52392	.47608	.3982
.07	.52790	.47210	.3980
.08	.53188	.46812	**.3977**
.09	.53586	.46414	.3973
.10	.53983	.46017	.3970
.11	.54380	.45620	.3965

EXERCISE

R_exercise_9.R
students.csv

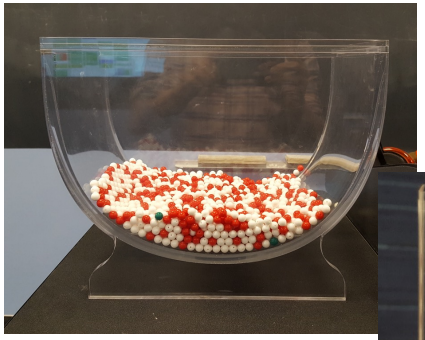
- Using the Normal Distribution and the z-scores we can go a first step beyond our data and calculate the probability that **a particular score** will occur **in a distribution**

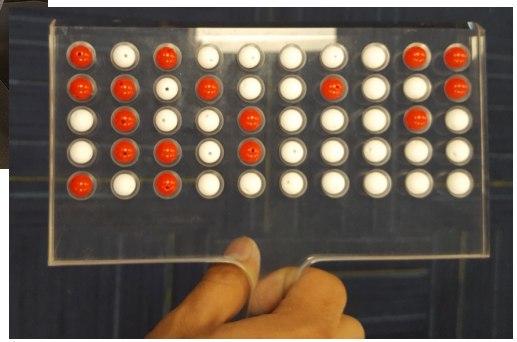
- **R_exercise_9.R** and **students.csv** (on moodle)
  - Plot the distribution of student heights
  - Plot the density of the data
  - Think about the scores as their z-scores (distance in SD from the mean)
    - What percentage of the students are less than 190 cm tall?
    - How likely is it that a student will be taller than 190 cm?

- **Homework**: how likely would you say it is that a student is between 160 and 190 cm?

Questions?

SAMPLING





SELECTING A SAMPLE

The Population

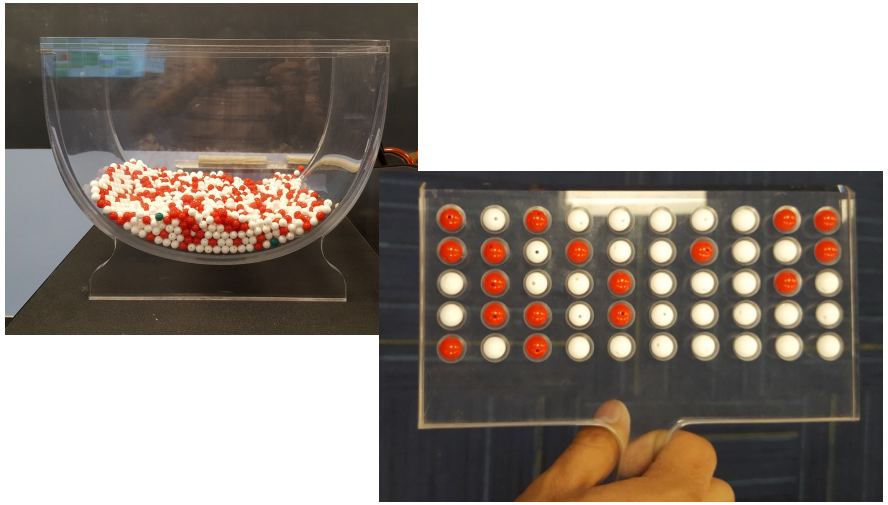
- includes all the individuals we are interested in
  (“women in Germany”, “words in the English language”, “people who had covid”)
- is typically too big to be observed directly

The Sample

- is typically much smaller than the population
- must be representative of the population for the inference to work
- should be selected at random

Tipton et al. Introduction to Statistics and Data Science
https://nustat.github.io/intro-stat-ds/index.html

SAMPLING FROM A POPULATION

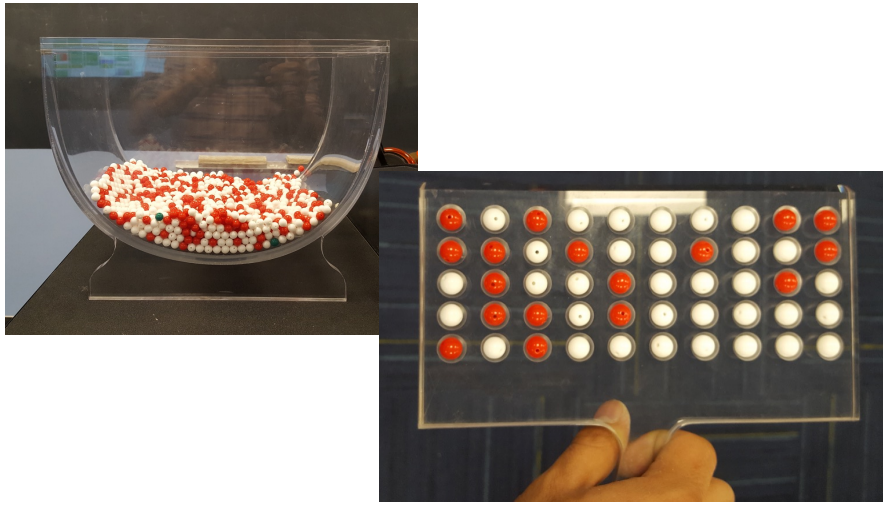


- How do we find out what percentage of these balls are red?
  - Is it enough to ***randomly select*** a sample (N = 50) and see how many are **red** (26%)?
  - What if we measured ***another, equally likely*** sample? Would our guess be the same?

- **Simulations** can give us an idea of what ***would*** happen ***if we did***

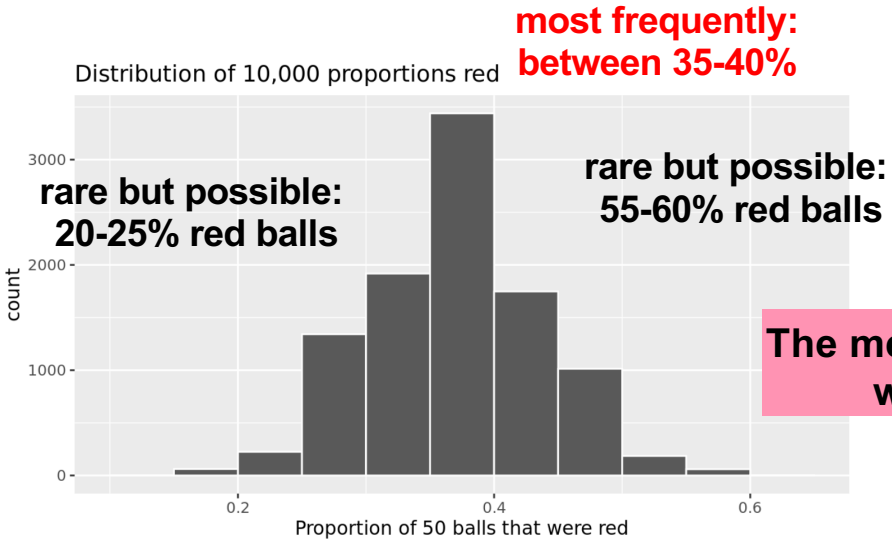
Tipton et al. Introduction to Statistics and Data Science
https://nustat.github.io/intro-stat-ds/index.html

SAMPLING FROM A POPULATION



- Let's imagine we take many samples from the same population (say, 10.000)
  - How do they look like?
  - Is the number of **red balls** always the same?

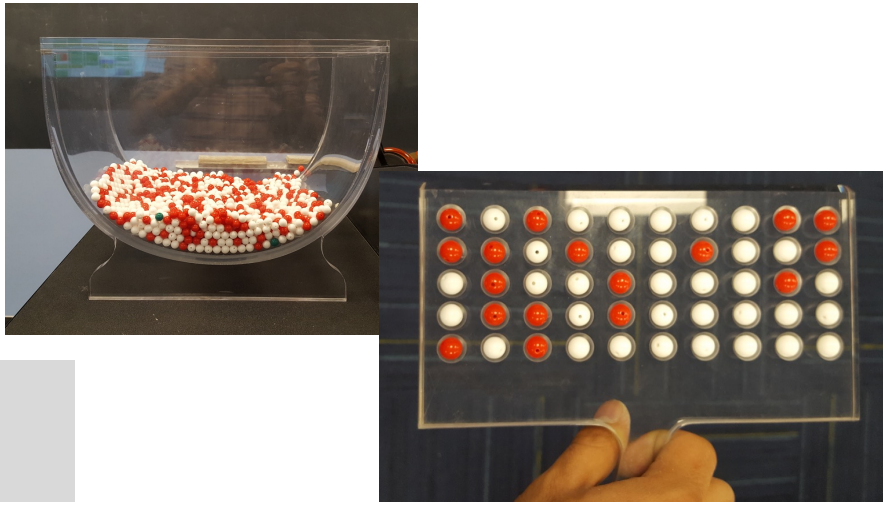
Variability due to chance



The mean of the sampling distribution will be the population mean

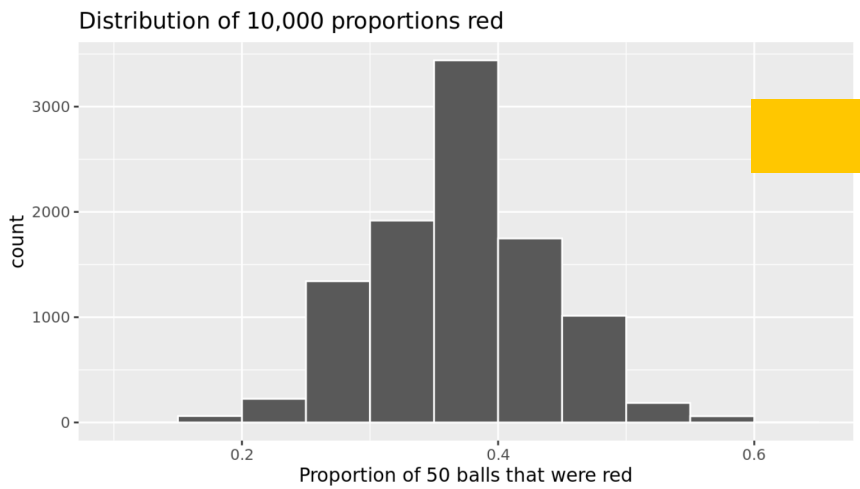
Tipton et al. Introduction to Statistics and Data Science
https://nustat.github.io/intro-stat-ds/index.html

SAMPLING DISTRIBUTIONS



A **Sampling Distribution**
is the distribution of a sample statistic
(mean, proportion, variance etc.)
in many samples from the same population

The **Standard Error**
is the SD of the sampling distribution
(how variable the distributions is / how different the samples are)



R_exercise_10.R

Tipton et al. Introduction to Statistics and Data Science
https://nustat.github.io/intro-stat-ds/index.html

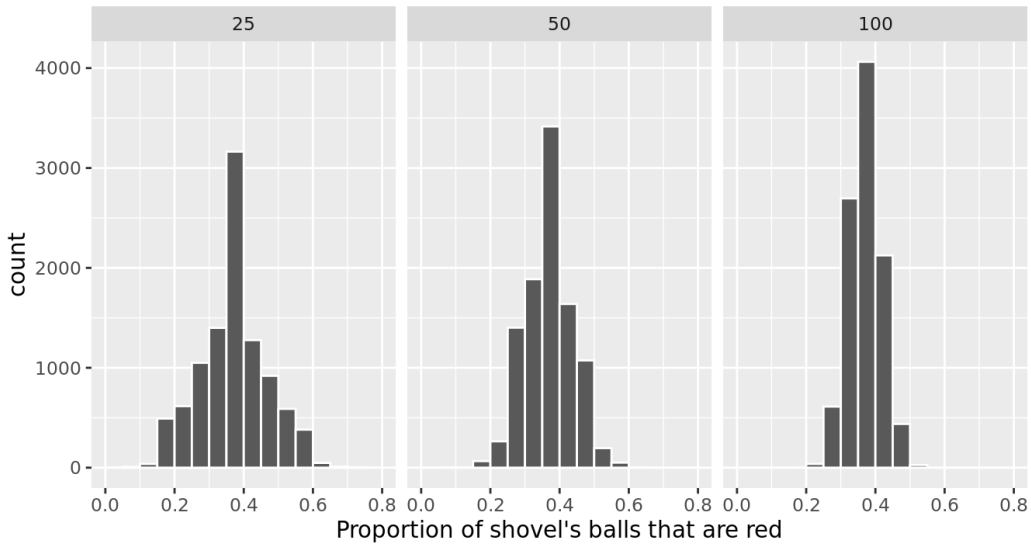
THE ROLE OF THE SAMPLE SIZE

↗ What happens if we increase the **size of the sample**?

1. the **sampling distribution** will get closer and closer to a normal distribution
2. the **spread** of the distribution (**standard error**) decreases
   - there is less variation in our sample proportions

Our guesses (based on the samples) get more consistent and precise

Comparing distributions of proportions red for 3 different shovels.

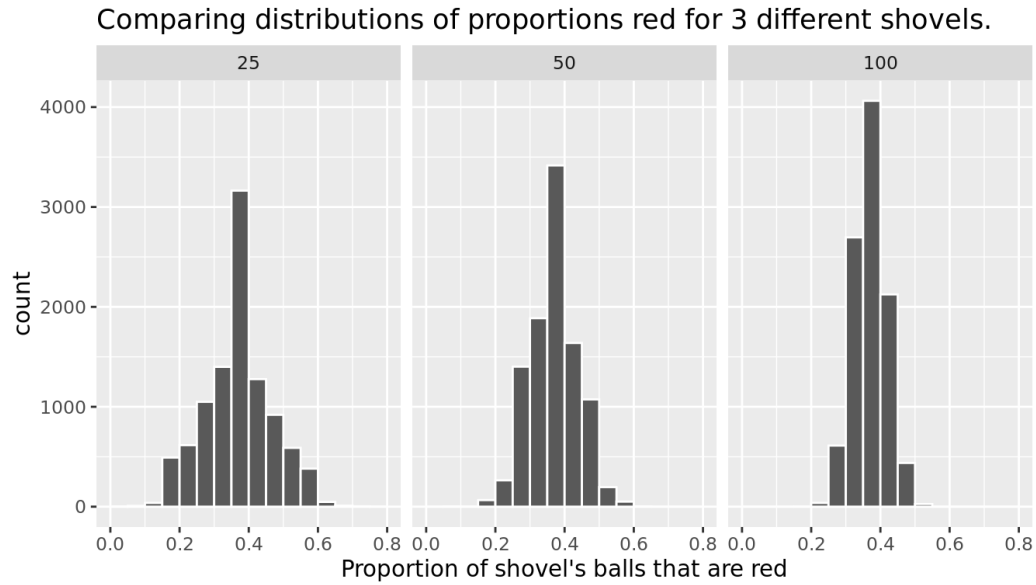


Tipton et al. Introduction to Statistics and Data Science
https://nustat.github.io/intro-stat-ds/index.html

THE ROLE OF THE SAMPLE SIZE

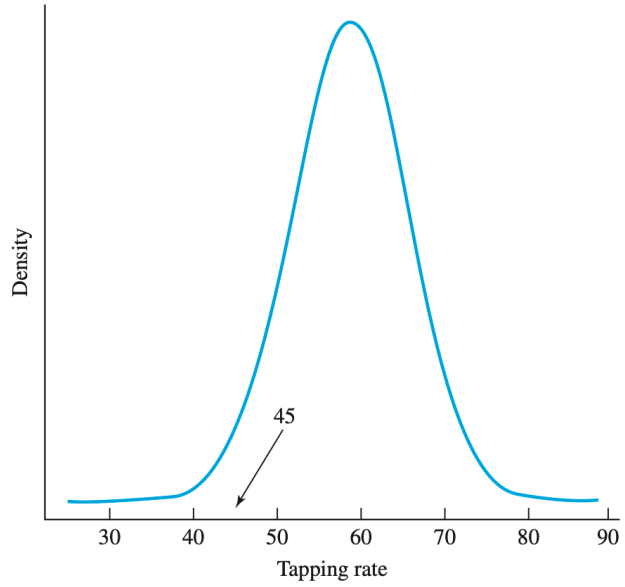
- **Central Limit Theorem**:
  given a population with mean μ and variance σ^2,
  the sampling distribution of the mean for a sample size N

  - will have a **mean** equal to μ
  - will have a **variance** equal to $\frac{\sigma^2}{N}$
  - will be approximated well by a **Normal Distribution** as N increases



Questions?

HYPOTHESIS TESTING WITH THE NORMAL DISTRIBUTION (SINGLE SCORES)



TALKING ABOUT HYPOTHESES

- In inferential statistics, we usually contrast two hypotheses
  - H_1 or **Alternative Hypothesis**: ***an effect is present***
    - Directional: also states the direction of the effect
      'students ***will know more*** about statistics after this class'
      before < after
    - Non-directional: an effect will occur, but it doesn't state the direction of the effect
      'students' knowledge of statistics ***will change*** after this class'
      before ≠ after
  - H_0 or **Null Hypothesis**: ***an effect is absent***
    'this class ***has not changed*** the students' knowledge of statistics'
    before = after

EXERCISE: STATE THE NULL HYPOTHESIS

- H_1: ***Your internet provider claims that the mean speed of your connection is > 3 Megabits/s***
  - H_0 (Directional):
    - the mean speed of your connection is < 3 Megabits

- H_1: ***A company claims that the mean entry level salary of IT professionals is > $58,000***
  - H_0 (Directional):
    - the mean entry level salary of IT professionals is < $58,000

- H_1: ***The mean time spent in jail by a first-time convicted burglar was 2.5 years, but now it has changed***
  - H_0 (Non-Directional):
    - the mean time in jail for a first-time convicted burglar is still 2.5 years

Howell, Fundamental Statistics for the Behavioral Sciences

TALKING ABOUT HYPOTHESES

- When a crime has been committed, and a person is accused of it, the suspect is considered ***innocent until proven guilty***
  - We convict the suspect only if
    - the **evidence** is too inconsistent with this belief
    - we can **disprove** the presumption of innocence
    - we believe that they are guilty beyond a reasonable doubt

"the null hypothesis is never proved or established, but is possibly disproven, in the course of experimentation"

R. A. Fisher

The Null Hypothesis is true until we manage to reject it

TALKING ABOUT HYPOTHESES

- We can't ***accept / prove*** H_1 using statistics, but we can ***reject*** H_0 or ***fail to reject*** H_0
  - Even if our data gives us confidence to ***reject*** H_0, then this may provide ***support*** for H_1 (but it ***doesn't prove*** H_1)

We can only talk about hypotheses in these terms!

FROM SAMPLE STATISTICS TO CONCLUSIONS ABOUT POPULATION PARAMETERS

population

sample

Inferential Statistics

- We observe a particular statistic (mean, difference between means, correlation) **in the sample**
- We want to know:
  - How likely is it that what we observe **in the sample** also happens **in the population**?
  - Or did I observe this in the sample just ***due to chance***?
- We need to find out:
  - What are the chances of obtaining this data if H_0 was true? $P(\text{result} \mid H_0)$
    - ***Only if they are low***: we reject H_0

AN EXAMPLE OF HYPOTHESIS TESTING

- H_1: there is a relationship between how students ***evaluate*** a course and the **grade they expect** to receive in that course
- H_0: there is no relationship between these two things

- We observe **a random sample** of 50 courses and find a general trend:
  - courses in which students expect to do ***well***: they tend to rate the quality as ***high***
  - courses in which students expect to do ***poorly***: they tend to rate the quality as ***low***

- We want to know:
  - What we saw in the data, is it the picture we expect to see ***in the population***?
  - If we ran the study again, would we find this result again, or ***was it due to chance***?
  - What are the chances of obtaining ***this data*** if the H_0 was true?

$$P(\text{result} \mid H_0)$$

Howell, Fundamental Statistics for the Behavioral Sciences

USING THE NORMAL DISTRIBUTION FOR HYPOTHESIS TESTING (SINGLE SCORES)

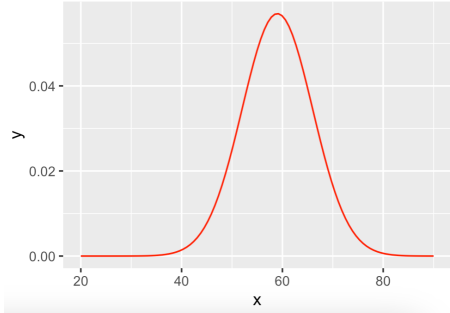
- Finger tapping rate can be used as a diagnostic tool for neurological disorders (Christianson & Leathem (2004)
  - Healthy adults: mean of 59 taps in 10 s (SD = 7)

  - We have just tested an individual who taps at a rate of 45 taps in 10 s
    - What could our Alternative Hypothesis (H_1) and our Null Hypothesis (H_0) be?
      - H_1 it doesn't belong to the healthy distribution of scores
      - H_0: it does

Howell, Fundamental Statistics for the Behavioral Sciences

USING THE NORMAL DISTRIBUTION FOR HYPOTHESIS TESTING (SINGLE SCORES)

1. How does the population of healthy adults look like?

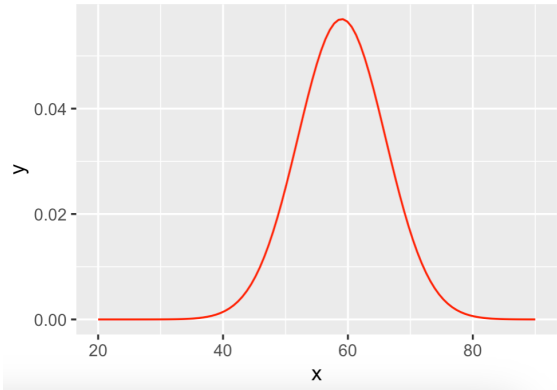


$\mu = 59$
$\sigma = 7$

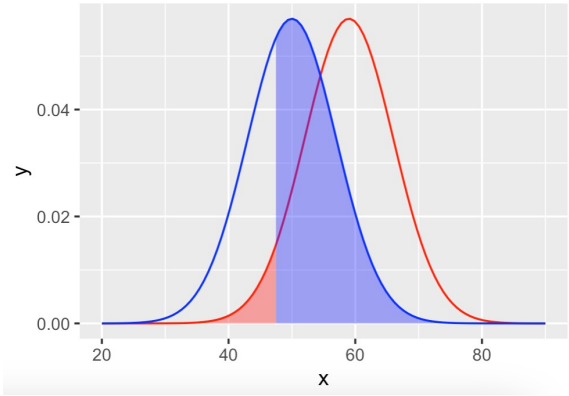
2. We know the rate of our patient $x = 45$

3. How likely is it that, if we picked a healthy adult ***randomly*** from the population, their rate would be ***as low as 45***?

"healthy adults"

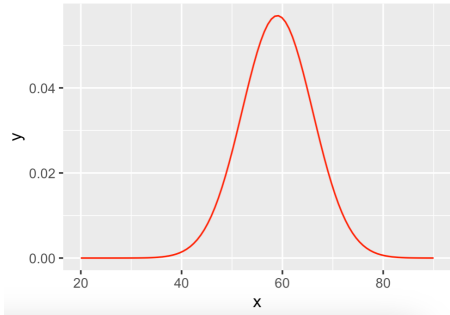


"adults with a pathology"



Howell, Fundamental Statistics for the Behavioral Sciences

USING THE NORMAL DISTRIBUTION FOR HYPOTHESIS TESTING (SINGLE SCORES)



$\mu = 59$
$\sigma = 7$

1. How does the population of healthy adults look like?

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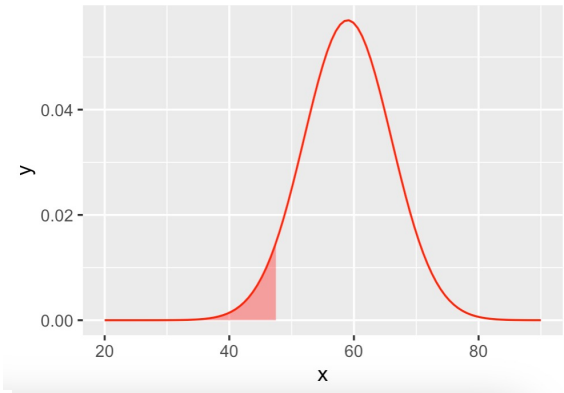
3. How likely is it that, if we picked a healthy adult ***randomly*** from the population, their rate would be ***as low as 45***?

R_exercise_11.R

"healthy adults"



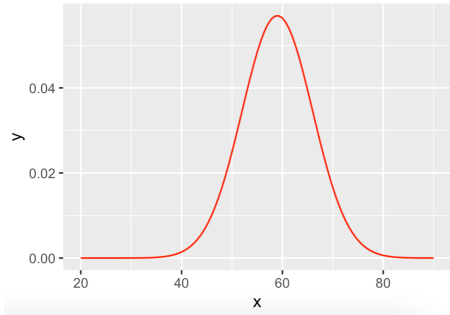
- **low probability**: reject H_0
- **not so low probability**: we fail to reject H_0

- It's the area below the curve for $x = 45$ in the normal distribution with $\mu = 59$ and $\sigma = 7$

$P(\text{score_this_low}|H_0) = .0228.$

Howell, Fundamental Statistics for the Behavioral Sciences

USING THE NORMAL DISTRIBUTION FOR HYPOTHESIS TESTING (SINGLE SCORES)



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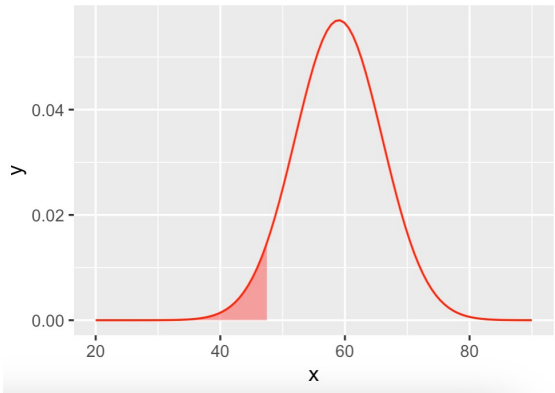
"healthy adults"



↗ **low probability**: reject H_0

↗ **not so low probability**: we fail to reject H_0

Is 0.0228 low enough tp reject H_0?
It depends!
(more on this in the next weeks)

$P(\text{score_this_low}|H_0) = .0228.$

SUMMING UP

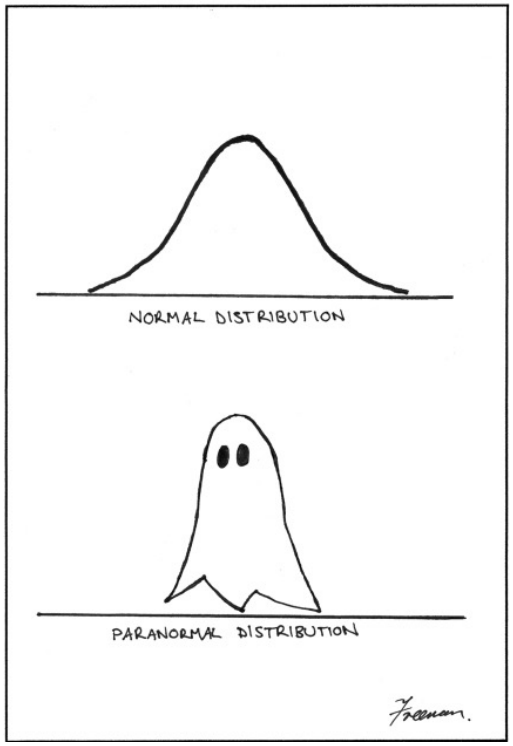
- The **Normal Distribution** can help us reason about the **probability** that a score comes from a distribution
  - the **area below the curve** can be interpreted as a probability

- In inferential statistics
  - We can only reject or fail to reject the **Null Hypothesis**
  - We collect **sample data** and compute the **sample statistics**
  - Depending on the **size of the sample**, are guesses about the **population** can be more or less consistent and precise

- To know if a score does not come from a population (**hypothesis testing on single scores**) we need to ask **how likely it is** that the score comes from the population
  - we reject or fail to reject H_0 ***on the basis of this probability***

Questions?

HOMEWORK

Freeman, Matthew. “A visual comparison of normal and paranormal distributions.” *Journal of Epidemiology and Community Health.* 2006 January; 60(1): 6.

- Chapter 8, sections 8.1 to 8.6 of Howell, “Fundamental Statistics for the Behavioral Sciences” (available on Moodle)
- Watch this videos:
  - “The Shape of Data: Distributions” (Crash Course Statistics) https://www.youtube.com/watch?v=bPFNxD3Yg6U
  - “Z-Scores and Percentiles” (Crash Course Statistics) https://www.youtube.com/watch?v=uAxyl_XfqXk
  - “The Normal Distribution” (Crash Course Statistics) https://www.youtube.com/watch?v=rBjft49MAO8
- Exercise in slide 23